

BIGMART SALES DASHBOARD

LOCHAN SHARMA | DE CA-2

Region
All

Category
All

Year
All

Sales proxy (Item_MRP)
801.2K

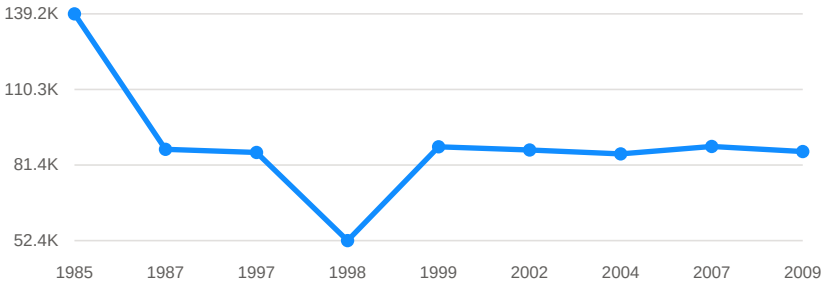
Rows
5,681

Average MRP
141.02

Top location tier
Tier 3

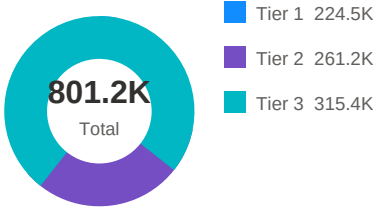
Monthly Sales Trend

Shown by outlet establishment year because month/date is absent



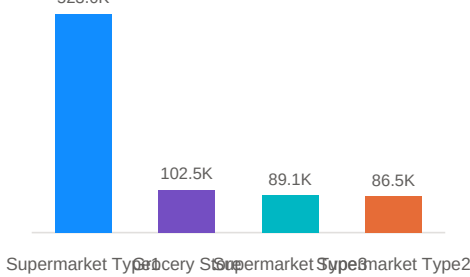
Sales by Region

Outlet_Location_Type



Sales by Outlet Type

Sales proxy by outlet



Profit by Category

Profit not available: average MRP used as category performance proxy



Dashboard Notes

Data limitations from source CSV

- No Date or Month column is present, so true monthly seasonality cannot be measured.
- No Profit, Cost, Quantity, or Item_Outlet_Sales column is present.
- Charts use Item_MRP as a transparent value proxy for dashboard practice.
- Lowest average MRP category: Baking Goods (129.05).
- Lowest average MRP region: Tier 2 (140.72).

Part A

Important dataset note: The provided BigMart CSV has Item_MRP, Item_Type, Outlet_Location_Type, Outlet_Type, and Outlet_Establishment_Year, but it does not have Sales, Profit, transaction Date, Month, or Year of sale. Therefore, the dashboard uses **Item_MRP as the sales/value proxy**. Conclusions below are based only on the available data.

Q1. Identify Trend

The overall dashboard trend is **fluctuating**, not a smooth continuous increase. Using Item_MRP by Outlet_Establishment_Year, the highest total value appears for 1985 (139.2K), while later years move up and down. The average MRP also stays in a narrow range, so the visual suggests fluctuation rather than a strong sales rise.

Q2. Identify Seasonality

Seasonality cannot be identified from this CSV because there is no Month or Date column. The dashboard can show value by outlet establishment year, but it cannot prove that any month is consistently high or low. A monthly sales field over multiple years would be required for a proper seasonality answer.

Q3. Business Insight

Actual loss/profit cannot be calculated because no Profit, Cost, Quantity, or Sales column is present. Using average MRP as a category performance proxy, **Baking Goods** is the weakest category (129.05), and **Tier 2** is the lowest location tier by average MRP (140.72). Recommendation: collect actual profit and sales data, then improve assortment, pricing, and promotions for weaker categories/tiers before deciding whether they are genuinely loss-making.

Part B

Q4. Why is stationarity important before applying ARIMA?

Stationarity is important before applying ARIMA because ARIMA assumes that the series has stable statistical behavior over time, especially mean, variance, and autocorrelation. If trend or seasonality remains in the data, the model can produce unreliable forecasts. Differencing is used to make a non-stationary series closer to stationary.

Q5. What is the difference between ARIMA and SARIMA?

ARIMA models non-seasonal time-series patterns using autoregressive, differencing, and moving-average terms: $ARIMA(p, d, q)$. SARIMA adds seasonal components: $SARIMA(p, d, q)(P, D, Q, s)$. SARIMA is preferred when the data has a repeating cycle, such as monthly yearly peaks, weekly demand cycles, or holiday effects.

Q6. Case Study: E-Commerce Sales Forecasting

For the e-commerce case, SARIMA should be chosen. The data has an increasing trend and a repeating seasonal pattern where sales rise every November and December and then drop after December. ARIMA can handle trend after differencing, but SARIMA is better because it explicitly models the yearly seasonal cycle. The 'S' in SARIMA means **Seasonal**; for monthly data, the seasonal period is usually 12.

Dashboard field	Used as	Reason
Item_MRP	Sales/value proxy	No sales column in CSV
Item_Type	Category	Closest category field
Outlet_Location_Type	Region	Tier 1 / Tier 2 / Tier 3 location grouping
Outlet_Establishment_Year	Time proxy	No transaction month/date column